The Field Equations of the Screened Khronon Action, and a Verdict for Every Physics Class

Working note (PAPER70), v1.1, 2026-11-07 (v1.0 deposited as DOI 10.5281/zenodo.24040313; this version adds the kernel erratum of Section 6). AI-assisted research programme; not peer reviewed. Companions: PAPER67 (DOI 10.5281/zenodo.23968171, the law and its failure record), PAPER68 (DOI 10.5281/zenodo.23990944, the action) and PAPER69 (DOI 10.5281/zenodo.24025170, the closure record). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

Abstract

Gravity carries one extra scale, $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, $\kappa = \frac{1}{2}$ fitted (proved not derivable), realised by the v3s screened khronon action. It passes Cassini, PPN, GW speed, galaxy lensing and SPARC, and fails AGC 114905, Malin 1 and the canonical FG1 tilt. It also under-fills X-COP clusters, groups, bright M31 dSphs and Coma UDGs by amounts that vary about 100× between classes, after the IMF accounts for the ellipticals and most of KiDS. Separately, it fails the CMB-lensing shape, which needs cold extra gravity at z 3400. No cold fluid, fixed-temperature sea, thermal light fermion or single measured key tried so far supplies either. The next lane must find one covariant quantity that sets that amplitude (not just a switch) in today's hot systems and is non-zero in the early universe, or show the two failures need separate causes. *(This is the checked seed paragraph of 2026-11-06, verbatim.)*

Since the seed, a joint systematics budget per class also explains the bright M31 dSphs, and a symmetric admissibility cut leaves the Coma UDGs NOT A TEST, so the measured boundary of the law is the X-COP clusters, the groups (fragile) and the CMB-lensing shape; and a shock-entropy sector written into the action fails by the no-tuning rule, so every physics class on the record carries a label. This note writes down the action in the Legendre form every lane solved (correcting the form printed in PAPER68), its complete covariant field equations checked against direct variation, their 3+1 form and a convergent static solver. It then lists every gate with its frozen label, states the residue as measured numbers, and collects the three frozen tests that can kill the law. Four physical classes of extra ingredient were built and graded on both footings: a cold fluid, a warm fixed-temperature sea, a hot thermal fermion, and a shock-entropy sector written into the action. Each meets its own wall, and no candidate passes every gate. The most probable reading of the record is the deposited, scoped law plus separate per-class systematics, with the cluster, group and ultra-diffuse rows and the CMB wall left unexplained.

1 Conventions

Units $c = 1$ and $K = 4\pi G$ unless stated; restoring c: $K \to 4\pi G/c^2$, the prefactor of R is $c^4/16\pi G$, and $a_0 \to a_0/c^2$ inside W. Two footings are carried and never pooled: canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. κ is fitted; a Lean proof shows that it is not derivable from the programme's premises. A gate is closed when it carries a label fixed by criteria committed before the first graded number; every lane runs a mutation twin that must fail.

2 The action, in Legendre form

PAPER68 prints $S = \frac{1}{4K}\int\sqrt{-g}\,[R - 2\mathcal{J}(Y) + 2\mathcal{K}(Q)] - \frac{\gamma_L}{2}\int\sqrt{-g}\,[\max(D\cdot P, 0)]^2 + S_m$. Read literally, P then appears only in the stiffness, its own equation forces $D\cdot P$ to be constant, and P decouples from gravity. The form every lane has actually solved puts $\mathcal{J}$ in Legendre form (QX17):

$$\frac{\mathcal{L}}{\sqrt{-g}} = \frac{R - 2\Lambda + 2\mu_K^2(Q-1)^2}{4K} + P^m A_m - W(K|P|) - \gamma_L F(\Pi) + \lambda\, n^m P_m, \qquad W(p) = \frac{a_0^2}{K}\Big[-\frac{p}{a_0} - \ln\Big(1 - \frac{p}{a_0}\Big)\Big], \quad (1)$$

with $F(\Pi) = \frac{1}{2}[\max(\Pi, 0)]^2$, $\Pi = D_m P^m$ and $\gamma_L = KL_\star^2/2$, plus minimally coupled matter S_m. Maximising over P returns the printed $\mathcal{J}$ exactly: $\max_P[P\cdot A - W(K|P|)] = (a_0^2/K)[u - \ln(1+u)]$ with $u = |A|/a_0$, attained at $K|P| = a_0 u/(1+u)$ (sympy, residual 0). So at $\gamma_L = 0$, Eq. (1) is the printed action; with $\gamma_L > 0$, P is an independent field and Eq. (1) is the theory. **This is an erratum on PAPER68's printed form only; no number changes.**

Adopted stiffness (owner decision, 2026-11-07). The owner adopted the Π^6 blend as the action's stiffness. In Eq. (1) the raw kink $F(\Pi) = \frac{1}{2}[\max(\Pi, 0)]^2$ is replaced by AC21 iter_6's blend, $F(\Pi) = b^2 f(\Pi/b)$ with $b = \rho_{\rm DE}$,

the a_0 law's own scale, so there is no new constant. Near 0, f is a pure power, $f(q) = (122363904/44610575)\, q^6$ for $q \le \frac{1}{2}$ ($m = 6 > d+2 = 5$). On the layer it is a C^4 convex blend, and $f(q) = q^2/2$ exactly for $q \ge 1$, so $F = \frac{1}{2}\Pi^2$ for $\Pi \ge b$. Eq. (1) with this F is the operative action of this note; it supersedes PAPER68's printed raw form. AC2q is not adopted; if it is adopted later, it carries the same blend with $b = X$. The scope of v3s and every graded label are unchanged: the blend moves no failing gate (largest effect $\le 1.4 \times 10^{-4}$ dex). The field equations of the next section hold for any stiffness F with Π_+ read as $F'(\Pi)$ throughout and the stress-tensor term written $\Pi F'(\Pi)\, h_{mn}$; for the raw F this is $2F h_{mn}$, as QX17 certified. QX17D certified the general form against a direct variation, for a generic polynomial F and for the blend's pure-power branch (residuals $\le 7.1 \times 10^{-40}$); the shortcut that keeps $2F h_{mn}$ fails the metric equation by up to 14 %. Limits adopted with the blend: well-posedness is local in time; it is tested in spherical 3D only, and non-radial ($l \ge 1$) modes are untested; folds are unchanged; power 6 is not forced (the smallest integer with $m > d+2$, admissible, not unique). The AC16 row remains the label of the raw switch.

Definitions. τ is the khronon; $Q = \sqrt{-g^{mn}\partial_m\tau\,\partial_n\tau}$ ($= 1/N$ in unitary gauge); $n_m = -\partial_m\tau/Q$; $h_{mn} = g_{mn} + n_m n_n$; $l = \ln Q$; $A_m = n^n\nabla_n n_m = -h_m{}^n\partial_n l$; $D_m = h_m{}^n\partial_n$; $\Pi = \nabla_m P^m - P^m A_m$; $\Pi_+ = \max(\Pi, 0)$; $\hat{P} = P/|P|$; $\mathcal{D}_m = \partial_m\Pi_+ - \Pi_+\partial_m l$.

3 The field equations (QX17, CERTIFIED)

$$\text{(E1)}\quad n^m P_m = 0, \tag{2}$$

$$\text{(E2)}\quad (1+\gamma_L\Pi_+)A_m + \gamma_L D_m\Pi_+ = KW'(K|P|)\,\hat{P}_m, \qquad \lambda = (1+\gamma_L\Pi_+)\,n\cdot\partial \ln N + \gamma_L\, n\cdot\partial\Pi_+, \tag{3}$$

$$\text{(E3)}\quad \nabla_m\big[n^m\rho_Q - \lambda P^m/Q\big] = 0, \qquad \rho_Q = \frac{\nabla_m[(1+\gamma_L\Pi_+)P^m]}{Q} + \frac{\mu_K^2(Q-1)}{K}, \tag{4}$$

$$\text{(E4)}\quad G_{mn} + \Lambda g_{mn} = 2K\big(T^{\rm mat}_{mn} + T^{Q}_{mn} + T^{W}_{mn} + T^{P}_{mn}\big), \tag{5}$$

with

$$T^{Q}_{mn} = g_{mn}\frac{\mu_K^2(Q-1)^2}{2K} + \frac{\mu_K^2}{K}Q(Q-1)\,n_m n_n, \qquad T^{W}_{mn} = -g_{mn}W + KW'\,\frac{P_m P_n}{|P|},$$

$$T^{P}_{mn} = g_{mn}(P\cdot A) + 2P_{(m}\partial_{n)}l + (\nabla\cdot P)\,n_m n_n - 2\lambda\, n_{(m}P_{n)} + \gamma_L\big[-g_{mn}F + \Pi F'(\Pi)\,h_{mn} + h_{mn}P^r\mathcal{D}_r - 2P_{(m}\mathcal{D}_{n)}\big].$$

(E5) Matter is minimally coupled, $\nabla^m T^{\rm mat}_{mn} = 0$. The energy density seen by the foliation is $n^m n^n T_{mn} = (\mu_K^2/2K)(Q^2-1) + W + \Pi + \gamma_L F$. At $\gamma_L = 0$ the leaf part of (E2) inverts to $K|P| = a_0|E|/(a_0 + |E|)$, the Legendre map; Π_+ is C^1, so the equations hold in the weak sense across the switch surface.

Checks. Each equation is compared with a direct Euler–Lagrange variation of Eq. (1) on jet symbols (mpmath, 40 digits, three random configurations in 2+1 dimensions): P equation max rel 3.8×10^{-41}, khronon equation 1.8×10^{-41}, full stress tensor 1.2×10^{-40}; FRW reduction exact. The MUTATE twin (two flipped signs) fails at 0.17–0.78.

AC2q (candidate, not adopted; QX17A, CERTIFIED). The dark-energy scalar is the clock, there is no $\mathcal{K}(Q)$ term, and the switch acts on the khronon charge $\Sigma = \Pi + Q^2$: $\mathcal{L}/\sqrt{-g} = R/4K + X - V(\phi) + P\cdot A - W - \gamma\{F(\Pi + 2X) - (2X)^2/2\} + \lambda n\cdot P$, $X = Q^2/2$. Its P equation is (E2) with $\Pi_+ \to \Sigma_+$; its clock equation is $\nabla_m[n^m\rho_\phi - \lambda P^m/Q] = -V'(\phi)$ with $\rho_\phi = J_P/Q + Q[1 - 2\gamma(\Sigma_+ - Q^2)]$, which is $\Box\phi = V'$ at $P = \gamma = 0$; at $P = 0$ the stiffness stress vanishes identically, so the background is the canonical scalar's. The MUTATE twin fails at $0.23/0.39/7.3 \times 10^{-3}$.

4 3+1 form, reductions and solutions

ADM (QX17B). P and N carry no time derivatives, so the P equation and the lapse equation are constraints on each leaf. The P sector's momentum density vanishes on shell, so the momentum constraint is GR's; the tensor principal part is GR's, so $c_T = c$. The Lean kernel file for these steps has 12 theorems, standard axioms only, and its MUTATE twin fails. The lane as a whole is not certified: its numerical ADM row failed as frozen on a sign convention in the checker's off-shell momentum density, and its `solve_bvp` spherical solver did not converge.

Reductions. FRW with $P = 0$: $\bar{Q} = 1 + I_0/(2\mu_K^2 a^3)$ and a dust-like $\rho \propto a^{-3}(1 + w_0/a^3)$. Static, weak, $\gamma_L = 0$: $\nabla \cdot [\mu(|\nabla\Phi|/a_0)\nabla\Phi] + \mu_K^2\Phi = 4\pi G\rho_b$, $\phi = \psi$. Spherical, $\mu_K = 0$: the law $g_N = g^2/(g + a_0)$ (Lean). Static, $\gamma_L > 0$: the Cassini hedgehog inside $L_\star$. AC2q FRW: the lapse–polarization mixing is $(1 - z)$ with $z = \gamma\dot{\phi}^2/\bar{N}^2$, so eliminating P multiplies the induced lapse term by $(1 - z)^2$, not the legacy $(1 - 2z)^2$ (which is the exact factor of a different action). The UV speed is $c^2 = z/(1 - z + z^2)$, and $0 < c_g \le c_{\rm ph} \le 1$ holds for any $f \ge 0$, so FRW health stands (FRW reconciliation chain: sympy ADM reduction and Lean, 11 theorems, standard axioms only; MUTATE 3/3 fail). The P block couples to the lapse only. The static-limit offset is half the earlier value, bounded by 1.1×10^{-10}. Still open: the AC7 Q2 / AC10–AC15 post-Newtonian chain carries the legacy factor and has not been rederived.

Solutions (QX17C, CERTIFIED as a solver). A conservative finite-difference Newton solver with a decaying Bessel outer boundary converges in 1–6 steps on both footings, about 5 s in all. It reproduces the law at $L_\star \to 0$ (worst 3.9×10^{-7}), the solar hedgehog to 7.1×10^{-6} of $g_\star = GM/L_\star^2$, and a $10^{10}M_\odot$ galaxy on the law to 4.8×10^{-6}; full-GR minus weak-field is 4.1×10^{-6} (Sun) and 5.4×10^{-7} (galaxy). An isolated point Sun reaches only 0.71/0.67 of the law at $10^3 L_\star$, so whether a galaxy built from screened stars shows the law is open.

5 Every gate, with its frozen label

The full table, one row per gate, is closure_2026_11_06/CLOSURE_FINAL.md; PAPER69's verdict table is its deposited form and is carried unchanged. The rows below are the 2026-11-06 lanes that came after PAPER69.

Gate	Label	Lane
Equations and health		
Covariant field equations, v3s and AC2q	CERTIFIED vs direct variation	QX17, QX17A
Field equations with the adopted blend stiffness (general F)	CERTIFIED with $\Pi F'(\Pi)\, h_{mn}$ in the stress tensor; the $\Pi_+ \to F'$ shortcut alone FAILS the metric equation	QX17D
3+1 form, Lean kernels	Lean PASS; lane NOT CERTIFIED (numerical row, solver)	QX17B
Static spherical solver	CERTIFIED, both footings	QX17C
Π^6 repair, curved 3D spherical switch	PASS; adopted by the owner 2026-11-07 as the stiffness; non-radial modes untested	BL3
AC2q decelerating slow roots	HARMLESS, scoped; $\epsilon_\phi = 2$ NOT REACHED by any record history	EP1, EP2
Nonlinear fold, shock-capturing	NO VERDICT as frozen (viscosity check fails); no pair where AC2q regularizes	FS1
Nonlinear fold, viscosity-free reading	DIFFERS, both footings: the 80 clock-only pairs fold 1.2 % earlier (DIFF-TIME EARLIER, edge-carried); the soft and phantom pairs (160) read SAME-REACH; no row is REGULAR and no pair shows AC2q regularizing; all checks pass	FS2
Moving Sun	MS1, MS2: NO VERDICT, bracketed (largest time-mean on any rung 0.147 of the line). MS3: PASS (moving Sun, scoped), FINAL for the door, with no MS4; every cell, gate and bracket below the line, with the largest at 0.617 of it (G1 Mars, M490 REFLECT bracket). The PASS rests on the frozen time-spread classes (NOISE / OSCILLATORY), not on a converged extrapolation. Disclosed: the ultra runs used a different memory allocator setting (allocator only; the chunked solver reproduces bitwise)	MS1, MS2, MS3
Light-cone causality	FAIL (physical, far below data); no repair in the khronometric class	AC18, SV2
All-gates battery (static gates G1–G8; G9 CMB)		
Six candidates on G1–G8	NOTHING PASSES ALL STATIC GATES; every candidate fails the Coma UDGs by at least $3.97\,\sigma_U$	GB1
Cold fluid: $\mathcal{K}(Q)$ dust, amplitude freed	FAIL: passes the background and CMB lensing, fails G1–G7; cosmic capture 4.111 in every class	KD1
Cold component decaying late	FAIL: CMB-lensing χ^2 46.3/53.0 against 18.48 at the longest lifetime discs allow	LD1

Gate	Label	Lane
Hot thermal fermion, 11.1377 eV	FAILS 7 of 9 gates; CMB-lensing χ^2 166.0–194.2	HC1
Shock-entropy field (proxy)	FAIL: X-COP and groups; need falls outward while the measured adiabat rises	EN1
Shock-entropy sector in the action	FAIL by the no-tuning rule; static solutions isentropic (f $\leq$ 0.04)	SE1
The step as several causes		
Multi-cause step	MULTI-CAUSE IN PART: IMF explains the ellipticals and KiDS	ST1
Joint systematics budget per class	IN PART: ellipticals, KiDS and bright M31 dSphs EXPLAINED; X-COP +0.440, groups +0.232, Coma UDGs +1.120 NOT EXPLAINED (canonical medians); CMB SEPARATE	SE2
Environment, re-read	NO: no environment key passes; satellites need about 1.6× the phantom, Coma UDGs 4–46×	EV1
Disc-only exclusion	every direction DEAD; even a perfect (oracle) exclusion fails GB1	DX1
One action-native scalar for both walls (OQ2 kept by the owner)	NO VERDICT as frozen. At the frozen fallback $N_{\min} = 8$ the S1 reading is DEAD on both footings, every start: SPARC vs groups at $\log q$ 3.0–3.4, gap up to 0.497 / 0.483 dex. Web row YES: the CMB wall needs the law's phantom off on the web ($F_{\mathrm{web,max}}$ 0.00481 / 0.00435). SEPARATED NO, as frozen	OQ1, OQ2
Outer edge of the law from the switch	NO EDGE IN THE ACTION: for an isolated galaxy the static switch variable is $\Sigma/X = 2+\rho_{\mathrm{ph}}/\rho_{\mathrm{DE}} \geq 2$, so the phantom halo runs on until an external field ends it (the standard EFE radius, already graded)	outer-edge check
Owner-opened test lanes, 2026-11-07 (adopt nothing)		
Host-held dust with the measured EFE capping the wells	FAIL (clusters, 0/20 cells, both footings); with no dust the EFE moves the KiDS baseline misfit to +0.870	QX7 iter_6
Remnant tracing the hot gas	SHAPE FAIL, both footings: the missing mass is more concentrated than the hot gas	QX7 iter_7
Asphericity of hydrostatic masses in the law's field	NEGLIGIBLE (≤ 0.005 dex, wrong sign), both footings	QX32
Everywhere matrix (23 rows × 17 columns, every cell sourced)	NO ROW PASSES EVERY COLUMN. The minimal framework cover is three rows that contradict each other (KD1 dust, QX27 retention, a literature row); UDGs, causality with well-posedness, and a derived a_0 stay uncovered by every framework row. Two effective fields (BK, QUMOND β 2.5) FAIL GB1 G2–G7 on both footings; BK passes only the ellipticals (+0.123 / +0.103)	EW1
Law switched off on the single-stream web (khronon multi-stream key)	FAIL (stage 2, population phantom), both footings. KiDS is kept: all 59 points lie inside the law-on turnaround radius. But the phantom that the galaxy population must keep to hold KiDS sources CMB lensing at q_{pop} 0.936 / 1.079 of the dust target at z 1, against a line of 0.0354 (×26 / ×30; ×11 / ×12 self-consistent). This holds for any key that keeps the law on where KiDS needs it	WL1
Two-switch door (law switch plus charge-switched sector)	graded by WL1: FAIL; EW3 keeps the map row only, no graded number	EW3, WL1
Fresh search for a new local action	no local action found; C1, C2 and C3 all FAIL, both footings. C1, any phantom multiplier of an absolute field length (the first absolute-scale grade): KiDS shift +0.283 / +0.260 against 0.05. C2, phantom screened beyond a second length: the natural local form does not screen, and the web forces that length to 0.3 Mpc or less, where KiDS fails. C3, two static branches: the static operator is not elliptic (radial), and the groups fail	FA2

Gate	Label	Lane
Nonlocal, history and global actions	NO CANDIDATE SURVIVES, both footings. I1, causal memory of the phantom: FAIL on the web row (r 75.4 / 78.8). I3, formation-epoch a_0 memory: FAIL G2–G7 and the web row. I4, conserved phantom charge: FAIL on the web row, NO VERDICT on G1–G7. Disclosed by the lane: I1's MUTATE M1 did not bite as frozen, and peak RSS (1.87 GB) exceeded the frozen 1 GB line	FA3
AeST (v4) on the web, in hot wells and in discs	NO-GO FOR THE CLASS, both footings: one length $1/\mu$ cannot hold both the CMB-lensing amplitude ($\mu \geq 10\,\mathrm{Mpc}^{-1}$; r 1.037 / 1.045 at 10) and KiDS ($\mu \leq 0.063\,\mathrm{Mpc}^{-1}$); the gap is ×160 (Lean `window_empty`). X-COP and groups FAIL at every computed μ: static dust $\leq 0.03\,M_b$ against a need of 2.39 / 2.07 M_b. The beyond-hydrostatic gate fails (0 % of X-COP radii). Primary CMB FAIL (read). At host-compatible μ, SPARC, Cassini, c_T and lensing = dynamics PASS. Two missed known answers are stated by the lane; no label moves	EW2
Late-formed compact baryonic objects (10^2–$10^7\,M_\odot$) as the hot-system half; the CMB wall untouched	FAILS (rows), both footings: X-COP shape, groups shape, Bullet. The budget is not excluded (lower bound 0.054 / 0.050 of the BBN baryons). No single formation rule: every key is S1 DEAD (gaps 1.50–2.53 dex). The dynamics window shrinks to 10^2–$1.8 \times 10^2\,M_\odot$. Objects tracing the stars miss X-COP's shape (rms 0.1208 / 0.1138), while an HSE-traced profile passes, so the needed profile lies between the gas and the stars. Groups pass 11 / 9 of 20 (canonical) and 10 / 8 (alt) against 16 needed. Disclosed by the lane: the Bullet child process peaked at 1188 MB, over the 1 GB line	NA2
Last two FA1 doors: disformal bulk-motion selector (I5), defect network of the a_0 scalar (I7)	I5 DEAD BY REDUCTION, both footings: every scalar the disformal metric allows reduces to dead keys (γ, X, ρ, p); with matter on the disformal metric, GW170817 confines BX to $[-6.2\times10^{-15}, +1.3\times10^{-15}]$, so the selector varies by at most 6.8×10^{-20}; no monotone weight passes S1; web lensing r 72–86 against 1.072. I7 DEAD, both footings: with η set by a_0 (2.240 / 2.463 meV) the network forms only at z of about 9; with η free, scaling networks track the background (w 0.167 at z 3400); the network is at most 3.7×10^{-4} of the baryons at z 3400 and at most 1.3×10^{-8} in clusters; gauge monopoles are particle-like. Disclosed by the lane: KA7 restated in amendment A1 before the rerun (labels unchanged); MUTATE M2 and M4 bite only weakly; the first run peaked at 1.025 GB, the rerun at 0.858 GB	NA1
The a_0 law plus an 11.1377 eV thermal fermion (test lane; the no-particle rule relaxed for this lane only)	does NOT pass every EW1 column, both footings. Run as a consistency test it reproduces the literature: X-COP 6/7 consistent, $M_{\rm req}/M_b$ 1.8–3.4, using 32–62 % of the supply, phase space never binding; run as a prediction (HC1) X-COP fails. Groups FAIL (14/20 / 11/20 against 16; the law over-predicts the shell); ellipticals FAIL 0/7 on phase space; AGC 114905 and Malin 1 FAIL for any fermion; KiDS G6 −0.098 dex FAIL; primary CMB $\Delta\chi^2$ +2716 / +2599 FAIL; CMB lensing, Coma UDGs and the colour split FAIL. SPARC PASS (not robust); Cassini and c_T PASS only with v3s/AC2q as the law's action. Disclosed by the lane: MUTATE M1 (mass halved) has NO POWER on X-COP, and the X-COP loader holds 7 clusters, not 12	EW4

Gate	Label	Lane
KiDS isolated-lens kill of the late-heated window (QX7 iter_4), as a joint systematic budget	NO VERDICT as frozen (the 4× MUTATE cannot bite on cell C3); no cell softens, both footings. KILL HOLDS on C2 and C4, NO POWER on C1 and C3. The hot gas measured by the kinetic SZ alone accounts for the raw +0.15–0.20 dex excess at 100–300 kpc; the posterior excess of data over the law is −0.21 to +0.05 dex (canonical) and −0.25 to +0.01 (alt), and the implied line is 0.05–0.24 dex (canonical) and 0.00–0.20 (alt), against the window's 0.17–0.25. Disclosed: the lowest mass bin uses an extrapolated gas row, and canonical C4's hold rests on it	KX1
SE2's two fragile rows, re-graded	Coma UDGs NOT A TEST (2 of 9 admissible at pericentre $d/3$; no orbit on disk); groups NOT EXPLAINED (+0.259 / +0.221), fragile; the residue is X-COP + groups + CMB lensing	UA1
Late-heated window on the cluster gates (QX7 iter_4 cells C1–C4)	FAIL on every cell, C3 included, both footings. Every cell passes CMB lensing (χ^2 8.70–11.69), $f\sigma_8$ at $z < 1$ (R_f 1.000) and the forest at z 2–4, and fails X-COP's profile, the groups and the Bullet. C3: X-COP +0.324 / +0.287 (inner bin +0.48: what evaporates from galaxies evaporates from clusters too), groups 0–1 of 20, Bullet FAIL (the sector is a fluid at Mach 1.7–1.9, ram-stripped, and follows the gas). Disclosed: the KX1 narrowing to C3 arrived after the numbers (C3 was frozen and graded as frozen); the Bullet child peaked at 1.48 GB, over the 1 GB line	CE2
Residual surface: does any pair of measured variables carry the law's missing mass across classes?	NO VERDICT as frozen: the Newtonian-f MUTATE does not bite, because $g_{\rm obs}$ was on the frozen variable list (from the brief) and makes the law-off f an exact function of $(g_N, g_{\rm obs})$. Main reading NO PAIR, both UDG readings. The closest pair, $g_N + g_{\rm obs}$ (χ^2/N 0.433 / 0.440), is circular and fails canonical leave-one-out. The best pair without $g_{\rm obs}$, depth + σ (χ^2/N 1.483 / 1.434), leaves SPARC at z −4.49 / −3.87, X-COP at +3.82 / +3.48 and the UDGs at +3.46 / +3.48; without $g_{\rm obs}$ no pair passes leave-one-out after dropping any class. SPARC is GB1's 145 galaxies	RS3
The kicked carrier, re-graded (RESOLUTION, 2026-11-07)		
Does the action release a kick when its switch changes branch?	NO RELEASE TERM, both footings: no isolated host changes branch at virialisation or at the acceleration crossover (min $\rho_{\rm ph}/\rho_{\rm DE}$ +0.000507 / +0.000462), and the blend gives zero latent energy at the branch point. Any kick is a postulate, not the action's	CK2
September's kicked collisionless carrier at today's needs, v_k 500–650 km/s	FAILS (rows) at every v_k on both footings; the binding gate is the SPARC high-mass tail, where every galaxy with $v_{\rm flat} \geq 250$ km/s keeps the carrier. The kick v_k is hand-set	CK1
Cosmology under the kicked carrier	NO VERDICT (the CMB-lensing gate has no power); among gates with power, v_k 550 passes and 575 / 600 fail the forest	CK4
A cold component born clumped at the future nodes, never in galaxies	FAIL, both footings, every clump mass: the primary CMB tolerates a bias of only 0.99670–1.00142, while a peaks-born population carries 45–1866 (bias 1.2 / 1.5 / 2.0 give Δ +4.10e4 / +5.57e5 / +6.35e6); the forest allows at most 0.13–0.28 % of the component in clumps; the clumps' own Poisson power fails CMB lensing at 10^{14} / $10^{15}\,M_\odot$ (+75.9 / +239.8) and passes at 10^{13}	NC2
Do cluster-member early types carry the component?	UNRESOLVED by rule, both footings: members sit +0.011 [−0.044, +0.065] / +0.012 [−0.043, +0.068] dex from field early types at fixed stellar mass and dispersion, so at most about 16 % extra mass at $r_{\rm half}$; stripping cannot hide a residue; the KiDS isolated-lens control is clean; CLASH's lensing-only slope leans to an early clump's sign with every interval containing 0	NC3

Gate	Label	Lane
The born-clumped clump spectrum (a record row; NC2 closed the candidate)	FAILS (rows) on both footings, binding K3: hosts inside R_{500} hold 0.0133 of the cosmic baryons and so 0.0091 / 0.0078 of the cold component (0.025 / 0.022 in the most generous bracket); clumps would need 110 / 128 times the observed host abundance (40 / 45 generous); even the ceiling with every baryon in a host is 0.75 / 0.68. K5 also fails: the top-hat collapse in the law's field fits X-COP at rms 0.166 / 0.143 at best. A fitted mass trend passes (slope 0.6–2.3 / 0.8–2.2 per dex). Disclosed: the Schmidt test never passed ($\langle V/V_{\rm max}\rangle$ 0.403–0.465), and a factor 2–3 would still leave the shortfall ≥ 13	NC4
Rows touched by the kernel erratum	the CMB-lensing and web-lensing numbers of KD1, SE1, CE2, NA1, FA2, FA3, EW2, OQ1 / OQ2 and SV5 above were computed with the Limber kernel corrected in Section 6; their frozen labels are printed as frozen, and the corrected values are listed there	GL1, GL3

6 Erratum to v1.0: the CMB-lensing Limber kernel

The Limber integrator shared by the sector and web lanes (QX3 iter2, `limber_ratio`) omitted the lensing potential's k^{-4}. It weighted each redshift by an extra χ^{-4}, so the ratio $S(L)$ was dominated by $z \sim 0.02$–0.1. CK4 found it, the code was read independently, and GL2 CONFIRMED it with its own code against CLASS: with k^{-4} restored the Limber shape over CLASS's own spectrum is 0.98–1.006 at L 20–400, against 0.045–33 without it. GL1 (sector side) and GL3 (web side) re-ran every affected lane with the kernel corrected; re-inserting the error reproduces the committed numbers (to $\leq 4.5 \times 10^{-16}$ on the web side), and a wrong power (k^{-2}) is the mutation. WG1–WG6, LD1 and HC1 used separate pipelines and are untouched. **Moved labels are re-stated pending owner; no frozen label is rewritten here.**

Label (lane)	Old	Corrected	Reading
Wall 1, CMB side (KD1 G9a, SE1 arm (a))	μ_J 2.51×10^4	μ_J 631; the binding side is BBN (μ_B 8.89×10^6, unchanged)	HOLDS
QX3 iter2 (no window)	$\mu_{\rm min}$ 3.98×10^4 vs $\mu_{\rm max}$ 79.43	$\mu_{\rm min}$ 631, still above 79.43	HOLDS
QX3 iter3 (class no-go, 35 cells)	best $\Delta\chi^2$ 92.59, 0/35	best 2.78, 1/35 (β 0.80, μ_0 1259)	re-stated pending owner
QX7 iter_3 (no window, 31 cells)	closest 17.87, 0/31	31/31 inside the lensing line	re-stated pending owner
QX7 iter_4 (FAIL on clusters)	window 270 cells, clusters 0	window 857 / 857, clusters 0 / 857	HOLDS
QX10 iter_1 (NO VERDICT)	J1 $\Delta\chi^2_{TT}$ 4.30 / 1.05 / 77.2 / 3.47	$\leq 10^{-7}$ on all four cells	re-stated pending owner (PASS by its own rule)
CE2, CMB-lensing gate	PASS, χ^2 8.70–11.69	$\Delta\chi^2$ 0.000 on every cell; its mutation no longer bites	gate re-stated pending owner (NO VERDICT); CE2's FAIL on X-COP, groups and the Bullet is kernel-free and HOLDS
CK4, CMB-lensing gate	NO VERDICT	NO VERDICT	HOLDS
SV5 iter_1, web amplitude W1 (line 1.072)	r 74.74 / 89.36	32.53 / 38.66	HOLDS (FAIL)
SV5 iter_1, web shape W2	$\Delta\chi^2$ 0.019, PASS	33.5 / 34.1, FAIL	re-stated pending owner (stricter)
SV5 T0.2, latest z for a late Jeans length of 30 / 100 / 1000 Mpc	none / none / none	2.0 / 1.5 / 0.75	re-stated pending owner

Label (lane)	Old	Corrected	Reading
SV5 iter_3, $a_0(z)$-tied Jeans scale, H-native	M^* $\Delta\chi^2$ 876.7, FAIL	0.016; K-mono 0.52	re-stated pending owner (WINDOW-CONDITIONAL; growth and lensing gates only)
WL1, population phantom line	$q_{\rm pop}$ 0.936 / 1.079 vs 0.0354	identical (kernel-free)	HOLDS
NA1 I5, web r	72.2–89.4	27.4–38.7	HOLDS (FAIL)
FA3 I1, web r	75.39 / 78.77	19.12 / 18.75	HOLDS (FAIL)
FA2 C2, web cut-off $L_{\rm max}$	0.3 Mpc	1.0 Mpc; KiDS still fails there (-0.074)	HOLDS
EW2, web window	μ_w 10, gap $\times$160	μ_w 0.63 / 1.0, gap $\times$10 / $\times$16	HOLDS (window empty)
OQ1 / OQ2, web row	$F_{\rm web,max}$ 0.00481 / 0.00435	0.00798 / 0.00720	HOLDS (YES)
QX28 G6b	$\Delta\chi^2$ 140.1 / 70.0, FAIL	0.006 / 0.004, PASS	re-stated pending owner; G6 still FAILS on the kernel-free G6a
QX27, QX29, QX31	FAIL; PASS (A-web conditional); ROBUST	labels unchanged	HOLDS

What moves is late, small-scale suppression: with the kernel corrected, PR4's shape at $L \leq 400$ is nearly blind to it. No move opens a door by itself, because the cluster, group and KiDS rows that close those doors are kernel-free. Wall 1 holds on its BBN side; on the CMB side alone the galaxy-versus-lensing gap narrows from $\times$316 to $\times$7.9.

7 What an all-gates ingredient would have to supply

On top of the law, the extra gravitating mass as a fraction f of the baryons (KD1 diagnostic, canonical / alt):

Where	Needed f
SPARC discs	≤ 0.33 / 0.19
X-COP clusters	~ 4.2 / 3.8
groups	1.2–3.5 / 0.9–2.9
X-ray ellipticals	~ 0.9 / 0.8 (about half of it IMF)
dSph satellites	~ 1.7 / 1.3
Coma UDGs	~ 80
the CMB	cold at z 3400 ($w \leq 0.01$), lensing $\chi^2 \leq 18.5$ at $L \leq 400$

8 Four walls

1. **Cold fluid** (KD1, QX3, QX7). Every well captures it, discs included. The CMB and BBN need $\mu_K \geq 8.9 \times 10^6\,{\rm Mpc}^{-1}$; galaxies allow at most 79.
2. **Warm fixed-temperature sea** (QX12). Retention $\exp(\text{depth}/c_s^2)$ is too sharp; the temperature would have to track each host.
3. **Hot thermal fermion** (HC1, HF1, VW1). Hot enough to stay out of discs, it under-fills clusters, cannot fill dSphs or UDGs, and fails CMB lensing; cold enough for the CMB, discs capture it. No window.
4. **Shock-entropy sector** (SE1). A conserved, non-cooling sector heated by its own shocks either sits isentropic, and holds almost nothing, or needs a different entropy history for each class at the same entropy. Its conserved rest mass also caps a host near the cosmic share, against the ~ 80 the Coma UDGs need.

The common reason: an ingredient that tracks each host's virial state has to virialise with the host; if it is also cold early, nothing keeps it out of discs. And the CMB needs the extra gravity at $z \simeq 3400$, where nothing is virialised, so any key that is on only in today's hot systems is off exactly where the CMB needs it on.

9 The residue: the law's measured boundary

SE2 draws every systematic knob jointly per class (IMF, velocity-dispersion scale, distance, anisotropy, membership, 3-D position, tidal admissibility). The posterior step of the plain law, in dex (median, 95 % interval):

Class	canonical	alt	label
X-COP clusters	+0.440 [+0.372, +0.484]	+0.408 [+0.341, +0.451]	NOT EXPLAINED
groups	+0.232 [+0.078, +0.362]	+0.194 [+0.041, +0.324]	NOT EXPLAINED
Coma UDGs (9)	+1.120 [+0.541, +1.718]	+1.075 [+0.472, +1.673]	NOT EXPLAINED
X-ray ellipticals	+0.067	+0.050	EXPLAINED
KiDS early/late	+0.085	+0.084	EXPLAINED
bright M31 dSphs (21)	+0.228	+0.219	EXPLAINED
CMB-lensing shape	cold extra gravity at z 3400; no key passes the hot classes		SEPARATE

The groups row is the weakest (z 3.04 / 2.55). Gas in the Coma UDGs is not in the budget and would lower their step; hydrostatic bias is set to 0 and would raise the cluster and group steps. These numbers are the boundary of the law as measured, not a mechanism. One profile meets both the inner-core requirement and the gas-to-stars band: a cored isothermal with r_c 50–100 kpc passes NA2's per-bin rule for 7/7 clusters on both footings (r_c 100 kpc: rms 0.086 / 0.086), though no collapse model in the record produces it (NC4, reported).

Update (UA1). UA1 re-graded SE2's two fragile rows. *Coma UDGs: NOT A TEST* on both footings: with tidal admissibility taken at pericentre $d_{\rm mean}/3$ in the Coma field, 2 of 9 are admissible. No orbit or proper motion exists on disk for any of the 9, so the label rests on the $d/3$ convention; with pericentres at $d/2$ or wider, 5 or more would be graded (not computed). NOT A TEST says the class cannot show it is in equilibrium; it does not say systematics explain the step. *Groups: NOT EXPLAINED* under the full single-width budget, +0.259 [+0.052, +0.459] (canonical) and +0.221 [+0.009, +0.415] (alt). The non-thermal term raises the step; distance and clumping lower it by about 0.01 dex each; the cross-catalogue term is 0.064 dex; the Eckmiller group masses are PROVISIONAL. The row is fragile: the alt lower edge is +0.009, and its sign rests on the non-thermal term (the inverted sign drops the median by 0.088 / 0.087 dex). With the same cut the bright M31 dSphs stay EXPLAINED (15 of 25). **The residue is X-COP + groups + CMB lensing**, the group-and-cluster scale, not the cluster scale alone.

10 Kill tests, frozen

Three tests can kill the law with data not yet in hand. Each is frozen before the data; nothing else is pending.

1. **Gaia DR4 wide binaries: Newton.** v3s screens binaries, so it predicts Newtonian motion: the frozen pipeline returns γ_∞ 1.0019 / 1.0100 (canonical / alt), not separable from Newton and separable from plain QUMOND (1.1494 / 1.2081). The prediction is sealed (CQ2, SHA-256 1d65509dbf). Scored as the screened (medium-first) reading of the frozen DR4 rows (owner decision 413c1c8e24), v3s is excluded iff $\hat\gamma - 1 \geq 3\sigma$, with the gate $\sigma_{\rm sys} < 0.048$ / 0.058 and the R rule (KILL iff $R - 1 \geq \max(0.01, 3\sigma_R)$). At the registered error (N 30,000, $\sigma_{\rm sys}$ 0.020) v3s is EXCLUDED for $\hat\gamma$ in 1.083–1.23; at the realistic N of about 8,000 the edge moves up.
2. $a_0(z)$ **flat to** z **2.33.** a_0 tracks $\rho_{\rm DE}$ (CFG689 quantiles, BAO + SN, no ΛCDM input). No sample on disk can decide it: all 52 cells are BLOCKED, because each sample's common mode (0.30–0.60 dex) is 4–11× the largest $e_{3\sigma}$ (0.084 dex at z^* 2.33). It needs a proposal-class ALMA sample with a same-estimator $z < 0.1$ reference (AT2). It kills the law if, at a powered z^*, the rival is favoured or neither model fits (AT1's rule).
3. **Lensing = dynamics.** v3s predicts $M_{\rm lens}/M_{\rm dyn} = 1$ to $\leq 6.5 \times 10^{-5}$ at 310 kpc. KiDS isolated lenses give $A = 1.129 \pm 0.206$ (canonical) and 1.035 ± 0.187 (alt), PASS but with weak power (V3W W3) The kill line is the frozen grade: FAIL if the KiDS ratio sits more than $2\sigma_{\rm tot}$ from 1, or the matched-host ratio more than $3\sigma_{\rm tot}$.

The exact statistics and kill lines are collected in closure_2026_11_07/PREDICTIONS_2026_11_07.md (52d4f27840), checked by its own script.

11 Reading

Every physics class on the record now carries a label, and none passes every gate. The law, and the action within its scope, describe cold rotating discs and the Solar System. The ellipticals, KiDS and the bright M31 dwarfs are accounted for by measured systematics. X-COP and the groups are not; the Coma UDGs are NOT A TEST on the record's orbit convention; and the CMB-lensing wall stands on its own. The elimination argument (closure_2026_11_07/DEDUCTION_2026_11_07.md) runs as follows. Three facts are measured: something cold gravitates at recombination; galaxies hold almost none of it today; clusters hold it today. No time-independent ingredient carries all three, and every switch on the law is closed. A component whose state changes in time leaves two cells. The escape cell, a late kick above the galaxy escape speed, is closed by X-COP. The puff cell, a late heating that leaves a diffuse halo of about $4\,M_b$ around galaxies, is closed by KiDS once the hot gas is measured rather than assumed (KX1). Its evaporated-core form, C3, fails X-COP, the groups and the Bullet (CE2), so every form of the puff cell is closed. The escape cell, re-graded at today's needs, fails on every kick from 500 to 650 km/s (CK1), and the action supplies no kick (CK2). A component born clumped at the nodes fails the CMB's bias tolerance and the forest (NC2), and cluster-member galaxies are as clean as field galaxies (NC3), so the exhaustion covers initial conditions as well as mechanisms (DEDUCTION section 8). One structural fact frames all of this: every host inside R_{500} together holds about 1 % of the cold component the CMB needs (0.0091 / 0.0078; at most 0.025 / 0.022 in the most generous bracket), so nearly all of it, whatever it is, lies outside bound objects (NC4). The elimination is complete for every ingredient the record can build. The a_0 law with the adopted v3s action is the law of cold isolated gravity; the CMB's cold component, the cluster and group residue, and the absence of either from galaxies are three measured facts that no single ingredient in reach of the action connects. Which of the three gives way is not decidable on disk: the galaxy leg is now the firmest (KX1); the cluster leg rests on seven X-COP clusters and provisional group masses (UA1); the CMB leg is a fit under the footing background, not independent evidence (QX15). This is the best remaining answer the record supports, not a closed theory.

12 What is unified, and what is not

From closure_2026_11_07/UNIFICATION_STATEMENT_2026_11_07.md, section 3, verbatim:

> Gravity carries one extra scale, the acceleration of the dark energy, a0 = (1/2) c sqrt(G rho_DE), and one action, the screened khronon action with its adopted stiffness, realises it as a complete and certified law of cold isolated gravity from the Solar System to galaxy lensing and gravitational waves. The same relation predicts a constant a0 to z 2.33 and Newtonian wide binaries, which are its kill tests. Beyond the cold isolated regime the universe contains a cold gravitating component that the CMB requires, clusters hold and galaxies lack, and no term in reach of the action, nor any ingredient the record could build, unifies that component with the law; the one candidate that does so adds a postulate under test today. That is the whole of the unification the record supports: one scale, one action, one measured boundary, and a named missing identity.

Since that statement was written, the candidate it names has been graded: CK1 FAILS it on every kick, and CK2 finds no term in the action that releases one. The kernel erratum (Section 6) leaves wall 1 and every web row standing. $\kappa = \frac{1}{2}$ stays fitted and not derivable; the vacuum-response identity a principle would have to supply is named and missing.

13 What waits on the owner

The PAPER68 erratum for the Legendre form. The Π^6 stiffness repair, whose case for and against was written for the owner (PX1, closure_2026_11_07/PI6_BLEND_ADOPTION_CASE_2026_11_07.md), was adopted by the owner on 2026-11-07 (closure_2026_11_07/OWNER_DECISION_2026-11-07_ADOPT_PI6_BLEND.md). The labels the kernel erratum re-states (Section 6) wait on the owner. Still running, for v1.2: AK1 (the fold-release regime test) and F1A. Every other owner-opened test lane above has landed with its label; none adopts anything.

Method

Every lane froze its criteria in a commit before any script ran, and every lane has a MUTATE twin that must fail. Footings are reported apart. No ΛCDM inputs were used. Scripts and outputs are in the repository under closure_2026_11_06/ and closure_2026_11_07/ at the commits the lane RESULT files name. The supplement archive deposited with this note holds every Lean source and build output, the closure tables, and each lane's criteria and result files, at the commit recorded as git_head in PAPER70_supplement_manifest.json, together with this note's source and audit script.